

dpi—higher than all the rest, which only support 600 x 600. Its interpolated scan resolution is also higher than the rest.

The optical resolution of a printer or scanner refers to the real hardware capabilities of the device, defining the fineness at which it can print or scan. Interpolated Resolution is the software-enhanced resolution of the device, and the figure is always misleading—don't go by it.

Scanner Type

The Canon MF3110 and Samsung SCX-4100 have flatbed scanners, and the Samsung SF-565P have only a sheet-fed scanner. The rest have both types of scanners.

A flatbed scanner has a moving scanning head that scans a document placed on the glass plate, whereas in a sheet-fed scanner, the document moves past a stationary scanning head. The paper cannot move past the scanner head as fast as the scanning head can move parallel to the paper. As a result, flatbed scanners are faster than sheet-fed scanners.

Documents are likely to get stuck or crumpled in a sheet-fed scanner if not placed in proper alignment with the paper guide. An advantage, though, with sheet-fed scanners is that they are more compact.

ADF

The Canon MF3110 and Samsung SCX-4100 are the only MFDs without an automatic document feeder (ADF). An ADF is important because it allows you to place a stack of papers to be



Canon MF3110

scanned—the MFD proceeds to scan and/or copy until it reaches the end of the stack. This means time savings, and that human intervention is minimised.

WIA Compliance

The Samsung SCX-4521F and Xerox WorkCentre PE220 are Windows Imaging Architecture (WIA)-compliant. WIA compliance means that the device enjoys native support in Windows XP—which recognises such devices as soon as you plug them in, and don't need to install any drivers. You can then use the Scanner and Camera Wizard to easily transfer scanned

How We Tested

Our test bed comprised a Pentium 4 3.2 GHz processor plugged onto an MSI 875P Neo FISR2 motherboard with 1 GB of Corsair 400 MHz DDR RAM, a Gigabyte nVidia GeForce FX 5950 graphics card, and a 250 GB 7200 rpm SATA Maxtor hard disk.

The system was powered by an Antec NeoPower 480 W power supply. We used Windows XP with SP1 as the OS, and the system was loaded with the latest chipset and graphics drivers.

We also installed the USB patch for XP, and connected the devices on the USB 2.0 port for optimal performance. We used 100 g/sm paper from Berga (A4 size) for printing.

Features

Various features such as the duty cycle, input and output tray capacity, input buffer, maximum print resolution supported and so on were rated. For the scanner, the maximum scan resolution, bit depth, scanner type, ADF presence and capacity, etc., were noted.

In the copier and fax aspect, the ability to work as a standalone device, the maximum number of copies in multi-copy mode, memory for storing fax pages, etc. were noted.

Performance

We tested performance on two parameters—speed and quality. Since MFDs consist of at least three basic parts—the printer, scanner, and the copier—we tested their performance as below.

Printer speed and quality test: We used a text document to test the raw print speeds of the printer. The quality settings used were Normal and Best. To test their ability to handle various aspects of a regular document, we created a combi-document with black text interspersed with graphs and images. We clocked the time it took the printers to print this document in Normal and Best quality modes, and analysed the print quality.

Scanner speed and quality test: We tested the scanners for speed using an average of five previews of an A4 size image. This was done when the scanner was fresh out of the box, so as to incorporate the warm-up time. Scanners with a higher warm-up time didn't fare as well. We also tested the warmed-up scanners to negate any possibility that a scanner might scan faster than the others once warmed up. We imported the same A4 size image at 150 dpi and 600 dpi, in Adobe Photoshop CS, and scanned a full text document in B&W mode at 200 dpi.

The scanner then underwent a test to determine its colour-differentiating abilities—we used a Kodak IT8 card with different shades of colour, each a little different from the preceding one. Next, we tested the resolution card to figure actual scanning capabilities by checking for its native resolutions. We scanned the card at 300 dpi at threshold colour depth to see how many boxes it could differentiate in between the lines. Finally, we did an OCR test to check the scanner's ability to differentiate light and dark areas. A print article was scanned into ABBYY FineReader Pro 8.0 PE, and verified for wrongly-interpreted letters in the software. We then gave a percentile value to the devices.

Copier: We copied a text document in black and white and rated the MFDs on the speed and quality of the printouts.

How the awards are given

The scores from features, performance and price are given weightages suitable to the category. An overall score out of 100 is then calculated. The product that scores the highest here is adjudged the *Digit Best Buy Gold* winner for the product category. The second-highest scorer gets the *Digit Best Buy Silver* award.